

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH adt.fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.3743	0.3683	1.02×	0.3093	0.3576	0.86×
Compiler	9	8	0.1443	0.0677	2.13×	0.1609	0.1816	0.89×
DBQuery	15	13	0.1186	0.0960	1.24×	0.1376	0.1656	0.83×
DecisionTree	7	6	3.633	3.123	1.16×	—	—	—
DomTree	15	14	0.1456	0.0618	2.36×	0.0593	0.0812	0.73×
KDTree	17	16	0.1764	0.1420	1.24×	—	—	—
LinearListReduction	11	11	0.0317	0.0059	5.41×	—	—	—
List	2	2	0.1328	0.1118	1.19×	0.2543	0.3381	0.75×
MonoTree	3	2	0.0172	0.0219	0.79×	0.0545	0.0781	0.70×
ObjectGraph	5	4	0.0885	0.0912	0.97×	0.1432	0.1627	0.88×
PiecewiseFunctions	8	7	0.1444	0.1046	1.38×	0.2404	0.2630	0.91×
TernaryTree	5	3	0.0275	0.0250	1.10×	0.0831	0.0915	0.91×
Trie	10	9	0.0650	0.0465	1.40×	0.1516	0.1660	0.91×
ColorOctree	25	18	0.0930	0.0836	1.11×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
OctTree	0.98×	0.75×	0.51×	0.61×	0.11×	0.40×	4.00×
Compiler	1.49×	0.52×	2.78×	0.87×	13.06×	0.33×	7.49×
DBQuery	1.04×	0.47×	1.29×	0.47×	8.47×	0.33×	5.85×
DecisionTree	1.17×	0.59×	17.88×	3.55×	0.83×	3.83×	35.55×
DomTree	1.58×	0.57×	1.82×	0.81×	16.01×	0.24×	10.37×
KDTree	1.27×	0.77×	1.29×	0.93×	0.19×	0.86×	4.81×
LinearListReduction	6.92×	0.90×	5.59×	3.12×	14.57×	5.05×	10.90×
List	0.82×	0.80×	3.71×	0.47×	0.10×	0.56×	1.47×
MonoTree	0.97×	0.85×	0.28×	0.68×	0.20×	0.69×	1.25×
ObjectGraph	0.94×	0.75×	3.42×	0.84×	0.39×	0.70×	2.90×
PiecewiseFunctions	1.13×	0.60×	2.69×	0.62×	2.11×	0.57×	4.56×
TernaryTree	1.16×	0.79×	0.37×	0.45×	0.22×	0.51×	1.41×
Trie	1.14×	0.56×	1.68×	0.84×	1.89×	0.20×	6.21×
Geomean	1.29×	0.67×	1.83×	0.86×	1.09×	0.64×	4.80×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
OctTree	2.6e+09 /2.7e+09	8.2e+09 /1.1e+10	1.5e+07 /3.0e+07	5.0e+05 /8.2e+05	6.4e+04 /6.0e+05	1.0e+04 /2.5e+04	6.6e+07/ 1.6e+07
Compiler	8.9e+08/ 6.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 4.7e+06	2.9e+05 /3.3e+05	1.5e+06/ 1.2e+05	1.2e+04 /3.7e+04	5.1e+07/ 6.8e+06
DBQuery	9.3e+08/ 8.9e+08	1.1e+09 /2.2e+09	3.9e+06/ 3.0e+06	1.7e+05 /3.6e+05	8.1e+05/ 9.6e+04	7.9e+03 /2.4e+04	2.1e+07/ 3.5e+06
DecisionTree	1.8e+10/ 1.6e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.0e+07	1.3e+05/ 3.7e+04	1.7e+06 /2.1e+06	1.3e+05/ 3.4e+04	1.1e+09/ 3.1e+07
DomTree	8.6e+08/ 5.5e+08	1.2e+09 /2.2e+09	1.2e+07/ 6.7e+06	9.8e+04 /1.2e+05	1.1e+06/ 6.8e+04	7.7e+03 /3.2e+04	5.7e+07/ 5.5e+06
KDTree	8.9e+08/ 7.1e+08	3.1e+09 /4.0e+09	1.4e+07/ 1.1e+07	1.6e+03 /1.7e+03	2.1e+04 /1.1e+05	1.3e+03 /1.6e+03	4.5e+07/ 9.3e+06
LinearListReduction	1.6e+08/ 2.3e+07	9.0e+07 /1.0e+08	1.5e+06/ 2.7e+05	7.6e+02/ 2.4e+02	2.8e+05/ 1.9e+04	6.5e+02/ 1.3e+02	1.2e+07/ 1.1e+06
List	1.3e+09 /1.5e+09	4.5e+09 /5.6e+09	9.4e+06/ 2.5e+06	2.6e+05 /5.5e+05	2.0e+04 /2.0e+05	6.6e+03 /1.2e+04	4.9e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.5e+05 /9.0e+05	3.3e+04 /4.8e+04	4.5e+03 /2.2e+04	2.5e+03 /3.6e+03	2.2e+06/ 1.8e+06
ObjectGraph	8.2e+08 /8.8e+08	3.0e+09 /4.0e+09	9.6e+06/ 2.8e+06	2.1e+05 /2.4e+05	3.1e+04 /8.0e+04	1.1e+04 /1.6e+04	2.4e+07/ 8.4e+06
PiecewiseFunctions	1.3e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.5e+07/ 5.4e+06	2.8e+05 /4.6e+05	2.0e+05/ 9.5e+04	2.0e+04 /3.6e+04	4.4e+07/ 9.7e+06
TernaryTree	4.1e+08/ 3.5e+08	1.2e+09 /1.5e+09	5.3e+05 /1.4e+06	6.6e+04 /1.5e+05	4.6e+03 /2.1e+04	2.5e+03 /5.0e+03	3.4e+06/ 2.4e+06
Trie	6.6e+08/ 5.8e+08	1.2e+09 /2.2e+09	8.4e+06/ 5.0e+06	1.9e+05 /2.2e+05	1.7e+05/ 8.9e+04	7.2e+03 /3.5e+04	2.6e+07/ 4.2e+06

Table 4: Per-pass performance for OctTree, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers. Times are median per iteration (s); ± shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med±err	SoA med±err	Speedup
barnesHutPotential	F	11/16	31%	0.0423	0.0450	0.94×
clearFlags	M	15/16	6%	0.1566	0.1800	0.87×
countActive	F	10/16	38%	0.0355	0.0331	1.07×
countParticles	F	8/16	50%	0.0365	0.0412	0.89×
fmmPotential	F	12/16	25%	0.0976	0.0984	0.99×
scaleEnergy	M	16/16	0%	0.1527	0.1776	0.86×
sumEnergy	F	12/16	25%	0.0356	0.0364	0.98×
sumMass	F	10/16	38%	0.0337	0.0305	1.10×
paletteEntriesQuantized	F	13/25	48%	0.0448	0.0390	1.15×
quantizationErrorProxy	F	10/25	60%	0.0482	0.0446	1.08×
Total				0.6836	0.7259	0.94×
Geomean						0.99×

Table 5: Per-pass PAPI counters for OctTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.1e+08 /2.2e+08	8.2e+08 /1.0e+09	6.7e+04 /5.7e+06	6.1e+02/ 5.7e+02	1.6e+03 /1.1e+05	4.3e+02 /4.4e+02	6.3e+06/ 1.7e+06
clearFlags	M	3.7e+08 /4.5e+08	8.0e+08 /1.7e+09	1.0e+06 /3.1e+06	2.7e+05 /4.2e+05	1.0e+04 /5.0e+04	3.2e+03 /1.3e+04	5.6e+06/ 2.4e+06
countActive	F	1.8e+08/ 1.6e+08	5.8e+08/ 5.8e+08	9.9e+05/ 4.7e+05	6.8e+02/ 4.6e+02	4.2e+03 /1.5e+04	5.8e+02/ 2.9e+02	6.2e+06/ 3.7e+05
countParticles	F	1.8e+08/ 1.5e+08	5.7e+08/ 5.7e+08	8.4e+05/ 3.1e+05	7.2e+02/ 4.8e+02	4.2e+03 /8.1e+03	5.9e+02/ 2.4e+02	6.1e+06/ 1.1e+05
fmmPotential	F	4.9e+08 /4.9e+08	1.6e+09 /1.8e+09	2.3e+04 /2.4e+06	4.7e+02 /1.0e+03	2.4e+03 /5.6e+04	3.9e+02 /8.6e+02	4.5e+06/ 2.2e+06
scaleEnergy	M	3.6e+08 /4.4e+08	9.1e+08 /1.9e+09	1.7e+06 /2.5e+06	2.2e+05 /3.9e+05	1.1e+04 /3.7e+04	3.4e+03 /9.4e+03	5.6e+06/ 2.5e+06
sumEnergy	F	1.8e+08 /1.8e+08	7.7e+08 /8.9e+08	3.6e+05 /5.7e+06	6.5e+02/ 4.6e+02	2.0e+03 /1.1e+05	4.5e+02/ 3.4e+02	6.5e+06/ 2.2e+06
sumMass	F	1.7e+08/ 1.5e+08	5.7e+08 /6.3e+08	7.4e+05 /2.4e+06	5.0e+02/ 3.9e+02	2.6e+03 /4.5e+04	3.9e+02/ 2.6e+02	6.4e+06/ 1.3e+06
paletteEntriesQuantized	F	2.3e+08/ 1.9e+08	6.7e+08 /7.6e+08	5.1e+06/ 3.7e+06	6.0e+02/ 5.1e+02	1.2e+04 /9.0e+04	4.5e+02/ 3.9e+02	9.4e+06/ 1.1e+06
quantizationErrorProxy	F	2.4e+08/ 2.3e+08	9.7e+08 /1.2e+09	4.1e+06/ 3.4e+06	2.2e+02 /4.0e+02	1.5e+04 /7.6e+04	1.8e+02 /3.9e+02	9.2e+06/ 2.6e+06
Total		2.6e+09/2.7e+09	8.2e+09/1.1e+10	1.5e+07/3.0e+07	5.0e+05/8.2e+05	6.4e+04/6.0e+05	1.0e+04/2.5e+04	6.6e+07/1.6e+07

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
blockCountPass	F	2/9	78%	0.0125	0.0039	3.23 $\times$
branchStatsPass	F	2/9	78%	0.0115	0.0047	2.47 $\times$
castInstCountPass	F	3/9	67%	0.0178	0.0021	8.36 $\times$
has cycle	F	4/9	56%	0.0345	0.0330	1.05 $\times$
instCountPass	F	2/9	78%	0.0140	0.0044	3.17 $\times$
latencyModelPass	F	3/9	67%	0.0118	0.0035	3.36 $\times$
memoryOpStatsPass	F	3/9	67%	0.0117	0.0059	1.99 $\times$
stripSideEffectsPass	M	7/9	22%	0.0800	0.0895	0.89 $\times$
targetReturnPass	M	9/9	0%	0.0809	0.0921	0.88 $\times$
throughputModelPass	F	3/9	67%	0.0118	0.0029	4.03 $\times$
verifyIR	F	9/9	0%	0.0184	0.0073	2.53 $\times$
Total				0.3052	0.2493	1.22 $\times$
Geomean						2.35 $\times$

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	5.2e+07/ 9.9e+06	5.1e+07/ 4.5e+07	1.3e+06/ 1.0e+05	3.1e+02/ 1.0e+02	2.5e+05/ 2.6e+03	2.7e+02/ 8.5e+01	4.7e+06/ 1.6e+04
branchStatsPass	F	5.8e+07/ 1.6e+07	7.9e+07 /8.6e+07	8.1e+05/ 2.0e+04	2.0e+02/ 1.2e+02	5.1e+04/ 3.4e+03	1.5e+02/ 9.5e+01	4.7e+06/ 4.8e+05
castInstCountPass	F	9.1e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.3e+04 /3.1e+04	1.6e+02/ 7.1e+01	8.6e+02 /1.1e+03	1.3e+02/ 5.8e+01	4.7e+06/ 1.9e+04
has cycle	F	1.3e+08/ 9.0e+07	1.1e+08 /1.5e+08	2.4e+06/ 1.1e+06	8.5e+04/ 5.9e+04	1.1e+05/ 2.2e+04	1.9e+03/ 5.7e+02	5.0e+06/ 1.4e+06
instCountPass	F	4.2e+07/ 1.1e+07	5.6e+07 /5.7e+07	1.4e+06/ 1.0e+05	3.4e+02/ 1.3e+02	2.9e+05/ 2.6e+03	3.0e+02/ 9.7e+01	4.7e+06/ 3.5e+04
latencyModelPass	F	6.0e+07/ 1.3e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.5e+04	2.2e+02/ 8.4e+01	2.0e+05/ 4.0e+03	2.0e+02/ 6.2e+01	4.8e+06/ 6.1e+05
memoryOpStatsPass	F	5.9e+07/ 1.7e+07	8.5e+07 /9.1e+07	6.6e+05/ 1.7e+05	3.8e+02/ 9.5e+01	3.1e+04/ 7.9e+03	3.6e+02/ 7.8e+01	4.7e+06/ 4.6e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	2.0e+06/ 1.1e+06	1.0e+05 /1.4e+05	6.5e+03 /1.9e+04	2.6e+03 /1.2e+04	4.3e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /2.1e+08	1.7e+08 /6.3e+08	1.2e+06 /1.2e+06	9.8e+04 /1.3e+05	5.9e+03 /2.4e+04	5.2e+03 /2.3e+04	3.7e+06/ 1.4e+06
throughputModelPass	F	6.0e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.2e+04	2.8e+02/ 1.2e+02	2.0e+05/ 3.6e+03	2.4e+02/ 8.7e+01	4.8e+06/ 6.8e+05
verifyIR	F	4.0e+07/ 1.6e+07	7.4e+07 /8.6e+07	1.1e+06/ 7.9e+05	8.2e+02/ 7.0e+02	4.1e+05/ 2.8e+04	7.0e+02/ 5.4e+02	4.7e+06/ 4.2e+05
Total		8.9e+08/6.0e+08	9.9e+08/1.9e+09	1.3e+07/4.7e+06	2.9e+05/3.3e+05	1.5e+06/1.2e+05	1.2e+04/3.7e+04	5.1e+07/6.8e+06

Table 8: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
clearQueryFlags	M	14/15	7%	0.0682	0.0819	0.83 $\times$
countJoins	F	3/15	80%	0.0186	0.0116	1.61 $\times$
filterSelectivitySkew	F	5/15	67%	0.0186	0.0124	1.50 $\times$
hashJoinPressure	F	5/15	67%	0.0190	0.0127	1.49 $\times$
scaleCosts	M	15/15	0%	0.0694	0.0837	0.83 $\times$
sumCost	F	6/15	60%	0.0186	0.0213	0.87 $\times$
sumMemory	F	6/15	60%	0.0184	0.0154	1.19 $\times$
sumRows	F	6/15	60%	0.0254	0.0227	1.12 $\times$
Total				0.2563	0.2616	0.98 $\times$
Geomean						1.14 $\times$

Table 9: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.6e+08 /2.1e+08	1.9e+08 /7.2e+08	2.9e+05 /1.4e+06	9.1e+04 /1.7e+05	2.9e+04/ 2.4e+04	2.9e+03 /9.0e+03	2.3e+06/ 9.3e+05
countJoins	F	9.4e+07/ 5.9e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.5e+04	3.6e+02/ 2.2e+02	1.4e+05/ 2.0e+03	3.3e+02/ 1.6e+02	2.7e+06/ 5.7e+04
filterSelectivitySkew	F	9.4e+07/ 6.2e+07	1.0e+08 /1.1e+08	5.1e+05/ 7.3e+04	2.0e+02/ 1.9e+02	1.2e+05/ 6.3e+03	1.8e+02/ 1.7e+02	2.7e+06/ 3.3e+05
hashJoinPressure	F	9.6e+07/ 6.4e+07	8.4e+07 /9.9e+07	7.3e+05/ 1.0e+05	3.9e+02/ 2.0e+02	1.9e+05/ 7.7e+03	3.6e+02/ 1.8e+02	2.7e+06/ 1.4e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.6e+05 /1.2e+06	7.4e+04 /1.8e+05	3.5e+04/ 2.6e+04	3.1e+03 /1.3e+04	2.5e+06/ 1.0e+06
sumCost	F	9.4e+07/ 7.4e+07	9.0e+07 /1.2e+08	5.6e+05/ 7.5e+04	5.4e+02/ 3.6e+02	1.2e+05/ 1.1e+04	4.0e+02/ 2.6e+02	2.7e+06/ 3.6e+05
sumMemory	F	9.2e+07/ 7.8e+07	9.0e+07 /1.2e+08	6.7e+05/ 8.0e+04	2.8e+02/ 1.6e+02	1.6e+05/ 1.1e+04	2.7e+02/ 1.4e+02	2.7e+06/ 3.7e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.9e+05/ 1.0e+05	2.8e+02/ 2.3e+02	2.2e+04/ 7.9e+03	2.7e+02/ 2.2e+02	2.4e+06/ 3.2e+05
Total		9.3e+08/8.9e+08	1.1e+09/2.2e+09	3.9e+06/3.0e+06	1.7e+05/3.6e+05	8.1e+05/9.6e+04	7.9e+03/2.4e+04	2.1e+07/3.5e+06

Table 10: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
classify Batch	F	5/7	29%	3.113	2.771	1.12 $\times$
classify Depth	F	4/7	43%	0.0447	0.0394	1.13 $\times$
classify tree	F	5/7	29%	0.0447	0.0397	1.13 $\times$
countFeatureUses	F	3/7	57%	0.0479	0.0318	1.51 $\times$
countLeaves	F	2/7	71%	0.0455	0.0251	1.82 $\times$
countNodes	F	2/7	71%	0.0459	0.0279	1.64 $\times$
countSmallLeaves	F	3/7	57%	0.0484	0.0327	1.48 $\times$
inferenceCost	F	2/7	71%	0.0485	0.0290	1.67 $\times$
sumImpurity	F	3/7	57%	0.0481	0.0323	1.49 $\times$
sumPathLengths	F	3/7	57%	0.0505	0.0338	1.50 $\times$
sumSamples	F	3/7	57%	0.0459	0.0319	1.44 $\times$
treeDepth	F	2/7	71%	0.0490	0.0280	1.75 $\times$
Total				3.633	3.123	1.16 $\times$
Geomean						1.45 $\times$

Table 11: Per-pass PAPI counters for **DecisionTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.4e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	1.1e+05/ 2.9e+04	1.4e+06 /2.0e+06	1.1e+05/ 2.7e+04	9.3e+08/ 1.8e+07
classify Depth	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.0e+05	1.5e+03/ 4.4e+02	2.1e+04 /2.8e+04	1.5e+03/ 3.7e+02	1.3e+07/ 3.3e+05
classify tree	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.0e+05	1.8e+03/ 5.6e+02	1.9e+04 /2.8e+04	1.7e+03/ 4.6e+02	1.3e+07/ 3.4e+05
countFeatureUses	F	2.4e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.7e+06/ 3.4e+05	6.9e+02 /7.0e+02	3.6e+04/ 1.4e+04	6.7e+02/ 6.6e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.3e+06/ 1.0e+05	1.2e+03/ 6.7e+02	4.5e+04/ 7.9e+03	1.2e+03/ 6.1e+02	1.4e+07/ 4.1e+05
countNodes	F	2.3e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.0e+06/ 1.1e+05	1.6e+03/ 9.4e+02	4.9e+04/ 7.9e+03	1.4e+03/ 7.8e+02	1.4e+07/ 4.4e+05
countSmallLeaves	F	2.4e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.9e+06/ 8.7e+04	8.4e+02/ 7.5e+02	3.3e+04/ 1.4e+04	8.2e+02/ 7.1e+02	1.6e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.2e+05	1.4e+03/ 7.5e+02	1.7e+04/ 7.8e+03	1.4e+03/ 7.0e+02	1.4e+07/ 4.1e+05
sumImpurity	F	2.4e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.9e+06/ 1.1e+06	1.4e+03/ 5.9e+02	3.6e+04/ 1.3e+04	1.4e+03/ 5.7e+02	1.5e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.7e+06/ 4.4e+05	2.8e+03/ 8.2e+02	3.9e+04/ 1.4e+04	2.7e+03/ 8.0e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.3e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.2e+06/ 2.5e+05	8.1e+02/ 5.6e+02	4.1e+04/ 1.4e+04	8.0e+02/ 5.4e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.5e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.1e+05	1.7e+03/ 6.3e+02	1.9e+04/ 7.9e+03	1.7e+03/ 5.8e+02	1.4e+07/ 4.2e+05
Total		1.8e+10/1.6e+10	4.0e+10/6.8e+10	5.4e+08/3.0e+07	1.3e+05/3.7e+04	1.7e+06/2.1e+06	1.3e+05/3.4e+04	1.1e+09/3.1e+07

Table 12: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
SumArea	F	6/15	60%	0.0379	0.0205	1.85 $\times$
computeWidths	M	14/15	7%	0.0302	0.0444	0.68 $\times$
count styled	F	3/15	80%	0.0360	0.0117	3.07 $\times$
find max Bottom	F	5/15	67%	0.0360	0.0185	1.95 $\times$
scaleLayout	M	15/15	0%	0.0290	0.0369	0.79 $\times$
sumTextWidth	F	3/15	80%	0.0358	0.0111	3.21 $\times$
Total				0.2049	0.1430	1.43 $\times$
Geomean						1.64 $\times$

Table 13: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.9e+08/ 9.9e+07	2.9e+08 /4.9e+08	2.5e+06/ 1.5e+06	8.3e+02/ 4.3e+02	1.7e+05/ 3.0e+04	6.8e+02/ 3.2e+02	1.4e+07/ 1.7e+06
computeWidths	M	6.6e+07 /1.4e+08	1.4e+08 /4.6e+08	1.3e+06 /1.6e+06	5.3e+04 /6.9e+04	5.4e+03 /1.0e+04	2.3e+03 /1.6e+04	1.3e+06/ 4.6e+05
count styled	F	1.8e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.3e+06/ 2.5e+04	7.5e+02/ 1.5e+02	3.0e+05/ 4.8e+03	7.1e+02/ 1.4e+02	1.4e+07/ 7.2e+05
find max Bottom	F	1.8e+08/ 9.4e+07	2.5e+08 /4.1e+08	2.5e+06/ 1.3e+06	5.3e+02/ 1.6e+02	1.5e+05/ 9.4e+03	5.2e+02/ 1.4e+02	1.4e+07/ 1.4e+06
scaleLayout	M	6.0e+07 /1.0e+08	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	4.3e+04 /5.1e+04	2.4e+04/ 9.2e+03	2.7e+03 /1.5e+04	1.4e+06/ 4.4e+05
sumTextWidth	F	1.8e+08/ 5.7e+07	2.0e+08 /2.4e+08	2.6e+06/ 1.7e+04	8.1e+02/ 1.5e+02	4.4e+05/ 4.3e+03	7.8e+02/ 1.4e+02	1.4e+07/ 7.3e+05
Total		8.6e+08/5.5e+08	1.2e+09/2.2e+09	1.2e+07/6.7e+06	9.8e+04/1.2e+05	1.1e+06/6.8e+04	7.7e+03/3.2e+04	5.7e+07/5.5e+06

Table 14: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
Find nearest Neighbour	F	13/17	24%	0.0274	0.0241	1.14 $\times$
countInRange tight_box	F	13/17	24%	0.0275	0.0197	1.40 $\times$
photonMappingBenchmark	F	12/17	29%	0.0418	0.0411	1.02 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0244	0.0164	1.49 $\times$
sumMassInRange	F	14/17	18%	0.0271	0.0202	1.34 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0282	0.0205	1.37 $\times$
Total				0.1764	0.1420	1.24 $\times$
Geomean						1.28 $\times$

Table 15: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.9e+06/ 2.2e+06	5.4e+02/ 4.6e+02	4.6e+03 /2.9e+04	4.0e+02/ 3.7e+02	7.8e+06/ 1.5e+06
countInRange tight_box	F	1.4e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.2e+06/ 2.1e+06	1.7e+02/ 1.5e+02	3.5e+03 /1.8e+04	1.5e+02/ 1.4e+02	7.8e+06/ 1.7e+06
photonMappingBenchmark	F	2.1e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.4e+06	3.2e+02 /5.3e+02	3.1e+03 /1.4e+04	3.0e+02 /5.2e+02	5.7e+06/ 1.5e+06
pointCloudNeighborhood	F	1.2e+08/ 8.3e+07	3.3e+08 /4.8e+08	2.3e+06/ 7.2e+05	1.6e+02/ 1.5e+02	4.1e+03 /1.7e+04	1.4e+02 /1.4e+02	7.9e+06/ 1.5e+06
sumMassInRange	F	1.4e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.3e+06 /2.7e+06	2.2e+02/ 2.0e+02	3.5e+03 /2.1e+04	2.0e+02 /2.1e+02	7.8e+06/ 1.9e+06
twoPointCorrelation bin_8_16	F	1.4e+08/ 1.0e+08	4.8e+08 /6.1e+08	2.1e+06/ 6.1e+05	1.6e+02 /1.8e+02	2.7e+03 /1.3e+04	1.4e+02 /1.7e+02	7.7e+06/ 1.3e+06
Total		8.9e+08/7.1e+08	3.1e+09/4.0e+09	1.4e+07/1.1e+07	1.6e+03/1.7e+03	2.1e+04/1.1e+05	1.3e+03/1.6e+03	4.5e+07/9.3e+06

Table 16: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/11	82%	0.0317	0.0059	5.41 $\times$
Total				0.0317	0.0059	5.41 $\times$
Geomean						5.41 $\times$

Table 17: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.6e+08/ 2.3e+07	9.0e+07 /1.0e+08	1.5e+06/ 2.7e+05	7.6e+02/ 2.4e+02	2.8e+05/ 1.9e+04	6.5e+02/ 1.3e+02	1.2e+07/ 1.1e+06
Total		1.6e+08/2.3e+07	9.0e+07/1.0e+08	1.5e+06/2.7e+05	7.6e+02/2.4e+02	2.8e+05/1.9e+04	6.5e+02/1.3e+02	1.2e+07/1.1e+06

Table 18: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
add1 List	M	2/2	0%	0.2543	0.3381	0.75 $\times$
length List	F	1/2	50%	0.0441	0.0215	2.05 $\times$
sumList	F	2/2	0%	0.0445	0.0452	0.99 $\times$
sumList tail recursive	F	2/2	0%	0.0442	0.0451	0.98 $\times$
Total				0.3872	0.4498	0.86 $\times$
Geomean						1.11 $\times$

Table 19: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.6e+08	1.9e+09 /2.8e+09	8.0e+06/ 1.3e+06	2.6e+05 /5.5e+05	1.1e+04 /5.5e+04	6.1e+03 /1.1e+04	9.7e+06/ 7.7e+06
length List	F	2.2e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.1e+05/ 9.9e+04	2.2e+02/ 1.1e+02	2.7e+03 /1.4e+04	1.9e+02/ 9.6e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	7.3e+05/ 5.4e+05	1.4e+02/ 9.8e+01	3.2e+03 /6.9e+04	1.2e+02/ 7.8e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.8e+05/ 5.7e+05	1.7e+02 /2.1e+02	2.9e+03 /5.8e+04	1.6e+02 /1.9e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.5e+09	4.5e+09/5.6e+09	9.4e+06/2.5e+06	2.6e+05/5.5e+05	2.0e+04/2.0e+05	6.6e+03/1.2e+04	4.9e+07/3.3e+07

Table 20: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
add1Tree	M	3/3	0%	0.0545	0.0781	0.70 $\times$
sumTree	F	3/3	0%	0.0090	0.0113	0.79 $\times$
sumTree TailRec	F	3/3	0%	0.0083	0.0106	0.78 $\times$
Total				0.0717	0.1000	0.72 $\times$
Geomean						0.76 $\times$

Table 21: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.3e+05 /8.4e+05	3.2e+04 /4.8e+04	3.0e+03 /1.4e+04	2.0e+03 /3.0e+03	5.3e+05/ 4.3e+05
sumTree	F	4.5e+07 /5.7e+07	2.2e+08 /2.6e+08	9.0e+03 /3.8e+04	2.5e+02 /2.7e+02	6.7e+02 /4.0e+03	2.2e+02 /2.3e+02	8.5e+05/ 6.8e+05
sumTree TailRec	F	4.2e+07 /5.4e+07	1.9e+08 /2.2e+08	5.7e+03 /2.3e+04	3.0e+02 /3.3e+02	7.6e+02 /4.2e+03	2.6e+02 /3.1e+02	8.6e+05/ 6.9e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.5e+05/9.0e+05	3.3e+04/4.8e+04	4.5e+03/2.2e+04	2.5e+03/3.6e+03	2.2e+06/1.8e+06

Table 22: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countLargeItems	F	3/5	40%	0.0122	0.0104	1.17 $\times$
countMarked	F	3/5	40%	0.0125	0.0137	0.92 $\times$
countSurvivors	F	4/5	20%	0.0133	0.0128	1.04 $\times$
deadBytes	F	4/5	20%	0.0131	0.0122	1.07 $\times$
liveBytes	F	4/5	20%	0.0128	0.0123	1.05 $\times$
sumObjIds	F	3/5	40%	0.0122	0.0101	1.20 $\times$
sweepUnmarked	M	5/5	0%	0.0722	0.0803	0.90 $\times$
totalHeapSize	F	3/5	40%	0.0124	0.0197	0.63 $\times$
touchHotObjects	M	5/5	0%	0.0710	0.0823	0.86 $\times$
Total				0.2317	0.2539	0.91 $\times$
Geomean						0.97 $\times$

Table 23: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.2e+07/ 5.2e+07	2.6e+08 /2.8e+08	1.2e+06/ 5.7e+04	4.6e+02/ 3.1e+02	3.4e+03 /3.9e+03	4.3e+02/ 2.7e+02	3.0e+06/ 7.2e+05
countMarked	F	6.3e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.2e+06/ 2.9e+05	4.9e+02/ 4.2e+02	3.5e+03 /8.2e+03	4.4e+02/ 3.8e+02	3.0e+06/ 6.6e+05
countSurvivors	F	6.7e+07/ 6.5e+07	3.1e+08 /3.8e+08	9.1e+05/ 5.8e+04	3.4e+02/ 2.7e+02	2.5e+03 /5.3e+03	3.1e+02/ 2.7e+02	2.9e+06/ 1.3e+06
deadBytes	F	6.6e+07/ 6.2e+07	2.7e+08 /3.5e+08	7.3e+05/ 4.5e+04	2.6e+02 /3.2e+02	2.3e+03 /5.3e+03	2.5e+02 /2.9e+02	2.8e+06/ 1.3e+06
liveBytes	F	6.5e+07/ 6.2e+07	2.7e+08 /3.5e+08	9.0e+05/ 4.7e+04	3.5e+02 /3.8e+02	2.5e+03 /5.6e+03	3.2e+02 /3.5e+02	2.9e+06/ 1.3e+06
sumObjIds	F	6.2e+07/ 5.1e+07	2.5e+08 /2.7e+08	1.1e+06/ 9.2e+04	3.2e+02/ 2.4e+02	3.1e+03 /4.3e+03	3.0e+02/ 2.1e+02	2.9e+06/ 6.9e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	1.1e+06/ 4.6e+05	9.7e+04 /1.1e+05	6.4e+03 /1.1e+04	4.7e+03 /7.1e+03	2.0e+06/ 9.9e+05
totalHeapSize	F	6.2e+07/ 5.6e+07	2.5e+08 /2.7e+08	1.2e+06 /1.3e+06	6.6e+02/ 6.5e+02	2.6e+03 /2.6e+04	5.2e+02 /5.2e+02	3.0e+06/ 5.1e+05
touchHotObjects	M	1.9e+08 /2.4e+08	5.7e+08 /9.0e+08	1.2e+06/ 4.8e+05	1.1e+05 /1.3e+05	4.8e+03 /1.1e+04	4.1e+03 /6.7e+03	1.8e+06/ 9.8e+05
Total		8.2e+08/8.8e+08	3.0e+09/4.0e+09	9.6e+06/2.8e+06	2.1e+05/2.4e+05	3.1e+04/8.0e+04	1.1e+04/1.6e+04	2.4e+07/8.4e+06

Table 24: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
addConstPW	M	8/8	0%	0.1193	0.1320	0.90 $\times$
autorefineMaxLevel	F	4/8	50%	0.0215	0.0198	1.09 $\times$
compressMass	F	3/8	62%	0.0204	0.0115	1.78 $\times$
diffPW	M	8/8	0%	0.1210	0.1309	0.92 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0223	0.0185	1.20 $\times$
norm2Estimate	F	5/8	38%	0.0263	0.0294	0.89 $\times$
pmapCutHistogram	F	4/8	50%	0.0333	0.0124	2.68 $\times$
truncateTolViolations	F	3/8	62%	0.0207	0.0130	1.59 $\times$
Total				0.3847	0.3675	1.05 $\times$
Geomean						1.28 $\times$

Table 25: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.4e+08	5.7e+08 /1.2e+09	1.5e+06 /2.0e+06	1.3e+05 /2.6e+05	1.3e+04 /2.2e+04	7.2e+03 /1.7e+04	4.5e+06/ 1.8e+06
autorefineMaxLevel	F	1.1e+08/ 1.0e+08	2.5e+08 /3.7e+08	2.5e+06/ 1.9e+05	7.8e+02/ 1.4e+02	6.2e+03 /6.3e+03	7.6e+02/ 1.3e+02	6.0e+06/ 9.3e+05
compressMass	F	1.0e+08/ 5.8e+07	2.5e+08 /2.8e+08	2.2e+06/ 3.3e+04	7.3e+02/ 3.5e+02	2.2e+04/ 5.2e+03	7.1e+02/ 3.4e+02	6.2e+06/ 6.9e+05
diffPW	M	2.8e+08 /3.3e+08	5.9e+08 /1.3e+09	8.0e+05 /2.0e+06	1.5e+05 /1.9e+05	9.7e+03 /2.1e+04	8.6e+03 /1.6e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.4e+07	3.3e+08 /4.9e+08	2.4e+06/ 2.5e+05	7.2e+02/ 1.8e+02	1.4e+04/ 1.0e+04	7.1e+02/ 1.8e+02	6.2e+06/ 1.6e+06
norm2Estimate	F	1.3e+08/ 8.9e+07	3.4e+08 /4.4e+08	1.4e+06/ 8.2e+05	7.1e+02/ 5.2e+02	1.1e+04 /1.7e+04	6.2e+02/ 3.8e+02	6.1e+06/ 9.9e+05
pmapCutHistogram	F	1.7e+08/ 6.3e+07	2.8e+08 /3.7e+08	1.7e+06/ 5.5e+04	8.8e+02/ 3.9e+02	1.1e+05/ 7.3e+03	8.5e+02/ 3.8e+02	4.7e+06/ 1.3e+06
truncateTolViolations	F	1.0e+08/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 3.1e+04	8.8e+02/ 4.6e+02	1.8e+04/ 4.8e+03	8.2e+02/ 4.4e+02	6.3e+06/ 6.9e+05
Total		1.3e+09/1.1e+09	2.8e+09/4.7e+09	1.5e+07/5.4e+06	2.8e+05/4.6e+05	2.0e+05/9.5e+04	2.0e+04/3.6e+04	4.4e+07/9.7e+06

Table 26: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-$ iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
add 1 tree	M	5/5	0%	0.0831	0.0915	0.91 $\times$
sum tree	F	5/5	0%	0.0275	0.0250	1.10 $\times$
Total				0.1107	0.1166	0.95 $\times$
Geomean						1.00 $\times$

Table 27: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.4e+05 /1.0e+06	6.6e+04 /1.5e+05	3.7e+03 /1.2e+04	2.4e+03 /4.9e+03	1.6e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.3e+08	4.6e+08 /5.4e+08	8.8e+04 /3.8e+05	1.3e+02/ 1.2e+02	9.1e+02 /9.0e+03	1.1e+02/ 1.0e+02	1.8e+06/ 1.3e+06
Total		4.1e+08/3.5e+08	1.2e+09/1.5e+09	5.3e+05/1.4e+06	6.6e+04/1.5e+05	4.6e+03/2.1e+04	2.5e+03/5.0e+03	3.4e+06/2.4e+06

Table 28: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-$ iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autocompleteTopKProxy	F	5/10	50%	0.0150	0.0094	1.60$\times$
countLazyNodes	F	4/10	60%	0.0128	0.0065	1.97$\times$
countTerminals	F	3/10	70%	0.0124	0.0101	1.23$\times$
decayTrieStats	M	10/10	0%	0.0752	0.0845	0.89 $\times$
resetTraversalState	M	10/10	0%	0.0763	0.0815	0.94 $\times$
sumPrefixFreq	F	3/10	70%	0.0126	0.0121	1.04 $\times$
sumSubtreeHints	F	3/10	70%	0.0122	0.0084	1.46$\times$
Total				0.2166	0.2124	1.02 $\times$
Geomean						1.25$\times$

Table 29: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.6e+07/ 4.3e+07	1.3e+08 /2.1e+08	1.4e+06/ 1.1e+06	1.4e+02/ 1.2e+02	5.6e+04/ 1.8e+04	1.2e+02/ 1.1e+02	3.8e+06/ 7.6e+05
countLazyNodes	F	6.5e+07/ 3.3e+07	1.3e+08 /1.6e+08	1.3e+06/ 2.7e+04	1.9e+02 /2.1e+02	2.0e+04/ 3.4e+03	1.8e+02 /2.0e+02	4.2e+06/ 6.6e+05
countTerminals	F	6.2e+07/ 3.1e+07	1.0e+08 /1.2e+08	1.2e+06/ 6.4e+05	3.1e+02/ 2.4e+02	2.6e+04/ 1.3e+04	2.7e+02/ 2.1e+02	4.1e+06/ 2.5e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	8.0e+05 /1.2e+06	9.4e+04 /1.2e+05	5.2e+03 /1.7e+04	2.6e+03 /2.3e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	1.7e+08 /2.0e+08	2.8e+08 /7.0e+08	1.1e+06/ 7.8e+05	9.1e+04 /1.1e+05	7.4e+03 /1.2e+04	3.3e+03 /1.2e+04	3.3e+06/ 9.0e+05
sumPrefixFreq	F	6.4e+07/ 3.0e+07	1.1e+08 /1.2e+08	1.2e+06/ 6.4e+05	5.8e+02/ 4.9e+02	2.8e+04/ 1.3e+04	4.6e+02/ 3.4e+02	4.1e+06/ 2.3e+05
sumSubtreeHints	F	6.2e+07/ 3.1e+07	1.1e+08 /1.2e+08	1.4e+06/ 6.4e+05	2.9e+02/ 1.4e+02	2.7e+04/ 1.3e+04	2.7e+02/ 1.2e+02	4.0e+06/ 2.8e+05
Total		6.6e+08/5.8e+08	1.2e+09/2.2e+09	8.4e+06/5.0e+06	1.9e+05/2.2e+05	1.7e+05/8.9e+04	7.2e+03/3.5e+04	2.6e+07/4.2e+06